

Experiment – 13
CSA1717 – Artificial Intelligence

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Aim:

To write a Python program to find the transpose of a given matrix.

Algorithm:

1. Start the program.
2. Read the number of rows and columns of the matrix.
3. Read the elements of the matrix.
4. Create an empty matrix for the transpose.
5. For each element at position (i, j) in the original matrix, place it at position (j, i) in the transpose matrix.
6. Display the transposed matrix.
7. Stop the program.

Source code:

```
# Program to find transpose of a matrix
```

```
# Taking matrix size
```

```
rows = int(input("Enter number of rows: "))
```

```
cols = int(input("Enter number of columns: "))
```

```
# Input Matrix
```

```
print("\nEnter elements of the Matrix:")
```

```
A = []
```

```
for i in range(rows):
```

```
    row = []
```

```
for j in range(cols):  
    row.append(int(input(f'A[{i}][{j}]: ')))  
A.append(row)
```

```
# Transpose Logic
```

```
T = []
```

```
for i in range(cols):
```

```
    row = []
```

```
    for j in range(rows):
```

```
        row.append(A[j][i])
```

```
    T.append(row)
```

```
# Display Transposed Matrix
```

```
print("\nTranspose of the Matrix:")
```

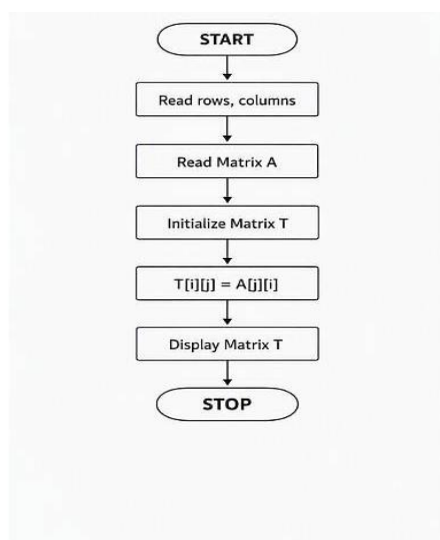
```
for i in range(cols):
```

```
    for j in range(rows):
```

```
        print(T[i][j], end=" ")
```

```
    print()
```

Flowchart:



Output:

```
File Edit Format Run Options Window H
Python 3.14.0 (tags/v3.14.0:ebf955d
Enter "help" below or click "Help"
>>>
= RESTART: C:/Users/hemal/OneDrive/
Enter number of rows: 2
Enter number of columns: 3

Enter elements of the Matrix:
A[0][0]: 1
A[0][1]: 2
A[0][2]: 3
A[1][0]: 4
A[1][1]: 5
A[1][2]: 6

Transpose of the Matrix:
1 4
2 5
3 6
```

Result:

The program successfully reads a matrix, computes its transpose by interchanging rows and columns, and displays the transposed matrix.